Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship. Explain what must be true for you to use it.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A cart starts at **3 m** and moves right at **5 m/s** for **2 s**. **Predict final position**. Allowed endpoints are **11** to **15 m**; **state this interval** and one thing you are treating as true.

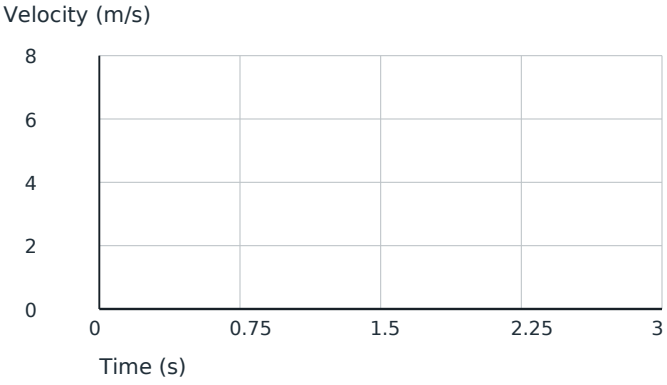
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v_f = v_i + a\,\Delta t$; $\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$.
Constant acceleration; i = initial, f = final. At constant velocity, $\Delta x = v\,\Delta t$.

Worked reasoning: From rest, a cart accelerates uniformly at **2 m/s²** for **3 s**. Final velocity=**6 m/s**; average velocity=(**0**+**6**)/**2**=**3 m/s**; displacement=**3**×**3**=**9 m**.

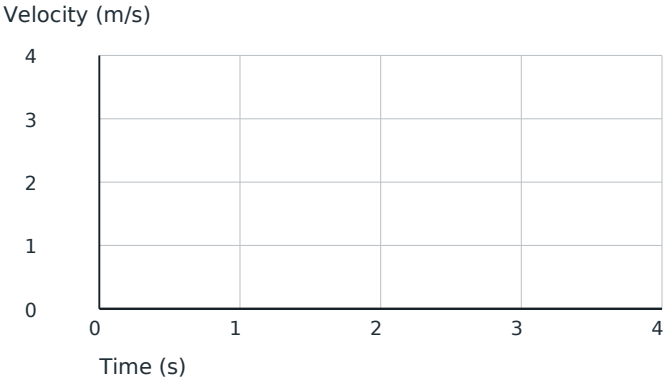
Your turn: Complete the cart's velocity-time graph. **Shade the triangular area under** the line and use it to **check the 9 m** displacement.



Marzano check - circle: ① ② ③

03 Choose the relationship

A robot drives straight at a steady **3 m/s** for **4 s**.
1. **Sketch its velocity-time graph**.
2. **Decide whether** a constant-velocity or acceleration relationship fits, then **find its displacement**.



Marzano check - circle: ① ② ③

04 Build final velocity

A remote-control car starts at **2 m/s** *east* and gains **2 m/s** each second for **3 s**. **Complete the table**. **How much** velocity does it gain altogether, and what is its final velocity?

Time (s)	Velocity east (m/s)
0	2
1	4
2	6
3	?

Marzano check - circle: ① ② ③

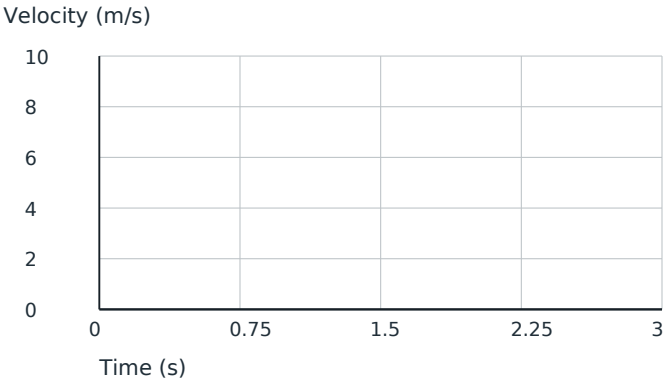
Choose a motion method

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Continue down each column.

05 Find the distance while speeding up

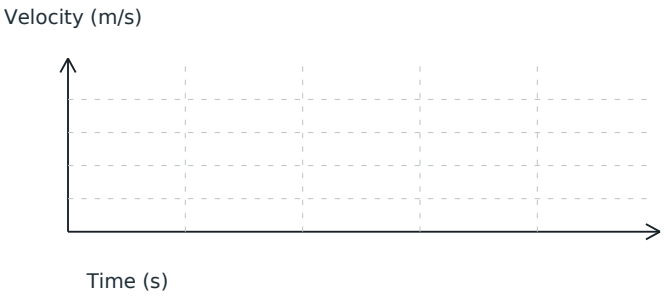
Use the car's velocities from question **4** to **complete the graph**. Its acceleration is constant. **Find its average velocity** and displacement over the **3 s**. **Use the graph**'s area to check.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

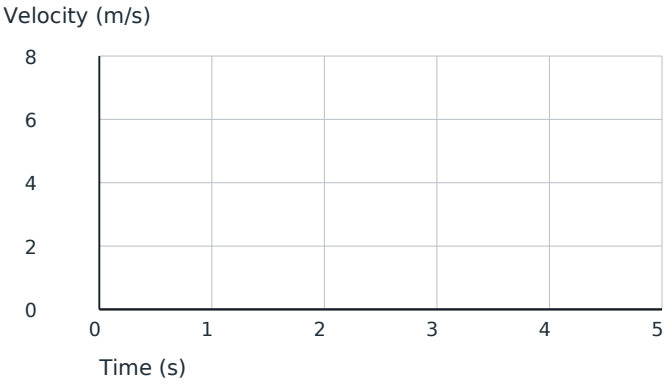
A cart travels *west* at **10 m/s** and *west* at **3 m/s 2 s** later. Is it speeding up? **Find acceleration size** and direction. **Sketch the velocity**–time graph; **label the given values**.



Marzano check - circle: ① ② ③

07 Exit check

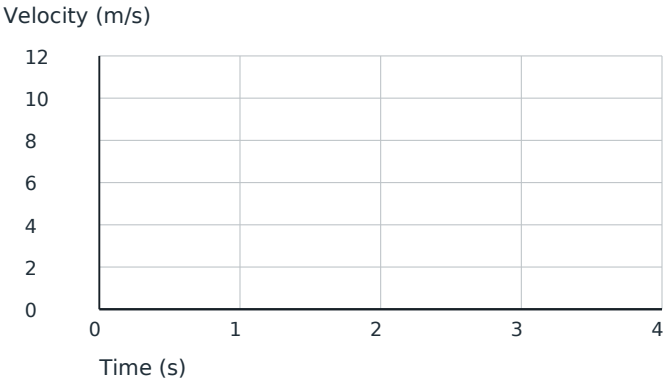
In a launch test, a small rover speeds up steadily from rest to **6 m/s** *east* in **5 s**. Plot the two endpoints and connect them. **Find acceleration**, average velocity and displacement.



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Create two different motions that travel **8 m** in **4 s**: one at steady speed and one speeding up uniformly from rest.
2. **Draw their velocity-time graphs**.



Marzano check - circle: ① ② ③